scientific reports

OPEN

Abnormal heart sound recognition using SVM and LSTM models in real-time mode

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Cardiovascular diseases are non-communicable diseases that are considered the leading cause of death worldwide accounting for 17.9 million fatalities. Auscultation of heart sounds is the most common and valuable way of diagnosing heart diseases. Normal heart sounds have a special rhythmic pattern as an indicator of heart integrity. Many experts concentrate on diagnosing the heart by automatic digital auscultation systems which find various distinguishable characteristics for heart sound classifications. This can decrease the mortality rate for cardiovascular diseases and enhance the patient's quality of life. This study aims to propose a real-time heart sound recognition system to classify both normal and abnormal phonocardiograms with the ability to define the abnormality type if existed. Digital signal processing methods, by applying the fast Fourier transform, filtering techniques, and the dual-tree complex wavelet transform, with machine learning classification algorithms are employed to segment the input phonocardiogram signal, extract meaningful features, and find the appropriate class for the input signal. We utilized three datasets, the PhysioNet of 1395, the GitHub of 800, and the PASCAL of 100 files segmented into three cardiac cycles. The proposed solution relies on the support vector machine and the long-short term memory neural network to distinguish between normal and abnormal heartbeat sounds and to recognize the type of abnormality (in the case distinguished) respectively. The results show that the proposed approach for normal/abnormal classification achieves an overall accuracy of 96.0 and 98.1%, sensitivity of 94.4 and 84.2%, and specificity of 64.9 and 98.4% for two and one support vector machines respectively among the state-of-the-art solutions. The long short-term memory model is also a well-known efficient classifier for temporal data, and the results show the accuracy of 99.2, 99.5, 98.6, and 99.4% for four (aortic stenosis (AS), mitral regurgitation (MR), mitral stenosis (MS), and mitral valve prolapse (MVP)), five (AS, MR, MS, MVP, and normal), six (AS, MR, MS, MVP, extrahls, and extrasystole), and seven classes (AS, MR, MS, MVP, extrahls, extrasystole, and normal). Furthermore, we found an efficient automatic segmentation method that was tested with the PASCAL database achieving a total error of 867,525.6 and 23,590.3 ms for datasets A and B respectively, with a computational time of 0.04 s to segment one cardiac cycle.

Keywords Deep learning, Heart sound, Long short-term memory, Machine learning, Phonocardiogram, Segmentation, Support vector machine, Classification.

Artificial intelligence (AI) applications have surged and invaded approximately all human-working fields due to their capability of achieving good results compared to humans. One of those vital sectors is the medical field, which is considered a sensitive sector having a direct impact on the enhancement of human health by either preventing occurrence of a health problem, finding the best treatment option for the patient, or even applying a medical intervention. There are several trends for AI applications in medicine, such as disease diagnosis, decision-making, drug development, treatment personalization, rehabilitative and surgical applications, and predictive practices[1].

This work is a combination of digital signal processing (DSP) and AI applications in the medical field for early abnormal heart sound recognition. Those sounds reveal important underlying structural cardiac physiological information which indicates pathological problems, either in the heart valves or in the heart muscle. The automatic classification of heart sounds has great practical significance for the screening and diagnosis of heart diseases. Early recognition of abnormality can help patients in getting immediate medical care, solve health problems in the early stages, enhance the quality of life, and reduce mortality[2].

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DSP is the process of analyzing, modifying, or extracting information from digital signals to optimize or improve their efficiency or perform certain tasks concerning time. The emergence and advancements in DSPs technology have significantly impacted several areas of science and engineering such as medical, space, commercial, military, scientific, industrial, networking, and telephone. Recently, DSP and AI are combined to obtain better processing results and to manipulate larger amounts of data, such as classifying sound signals including heart sounds.

The cardiac cycle is identified as a series of events that start from the beginning of a heartbeat and end at the beginning of the next beat. During this cycle, contractions and relaxations in the heart chambers, atria, and ventricles occur which are known as systole and diastole. The atria and ventricles alternate in systolic and diastolic operations, both cannot be in the same state simultaneously. Sounds are produced when the air around a subject vibrates. Some inner parts of the heart vibrate due to physiological or pathological causes, such as the normal mechanical operations of the closure of the heart valves, or pathological operations such as the change in the heart's size. Audible normal sounds are known as Fundamental Heart Sounds (FHS), they can happen at the beginning of the systolic or diastolic periods in every heartbeat. The FHS includes the first heart sound (S1) and the second heart sound (S2)[3].

In pathological cases, the heart produces abnormal heart sounds which can be an indicator of various diseases. The abnormal heart sounds include, the third heart sound (S3), the fourth heart sound (S4), murmurs, opening snaps (OS), rubs, and clicks. Cardiovascular diseases (CVDs) are types of Non-communicable diseases which considered the leading cause of death worldwide with 17.9 million deaths (32% of all global deaths) yearly based on the World Health Organization 2022 (WHO) statistics. CVDs account for 39% of deaths in Jordan and it is classified as the main cause of morbidity and disability according to the Jordan National STEPwise survey[4].

Modern health care and public health emphasize the importance of disease prevention either with primary, secondary, or tertiary methods. The secondary prevention of CVD aims to lower the effects of the disease by applying a vital step which is early diagnosis. When this step is performed the patient could avoid critical and permanent damage by receiving a suitable intervention at the right time[5]. Machine Learning and DSP can be used efficiently for recognizing Abnormal Heart Sounds and they can help in the early detection and classification of various types of abnormalities. Classical signal processing of heart sounds collects domain knowledge to acquire signal representations to be used in ML. Heart signals in DSP can be integrated directly with ML performing prediction, dimensionality reduction, or classification as a part of the system. In both cases, DSP and ML can analyze, extract features, and classify Heart Sounds to distinguish between normal and abnormal types[6].

Heart sound detection and classification have a significant interest in many pieces of research, which are intended to understand the nature of the signal and the major discriminative properties to distinguish between Normal from Abnormal heart sounds. The need for these studies is due to the importance of early detection of abnormality to avoid mortality or morbidity by enabling appropriate medical interventions, the recent advancements in medical wearable devices for continuous monitoring, and the evolution of computer skills.

Through the work on automatic heart sound classification, different methods have been introduced, where some of them depended on Classical approaches while others depended on End-To-End approaches. In our work, we are presenting a Classical approach, for features extraction, by combining FFT, dual-tree complex wavelet transform (DTCWT), the SVM classifier for Normal/Abnormal heart sound classification, and the Long-short Term Memory neural network (LSTM) classifier for abnormality detection up to six classes.

Literature review

The heart sounds recognition approaches can be divided into classical and End-to-End approaches as shown in (Fig. 1). Classical heart sound recognition approaches work as a pipeline of several stages which appear like a white box. In each step, human intervention is required to select a suitable algorithm or technique. The heart sounds recognition process involves stages for automatic analysis and classification. The process begins with data acquisition of the PCG signal which requires pre-processing. The second step involves segmentation of the pre-processed PCG followed by feature extraction and selection, and the final step is the classification of heart sounds.

Unlike classical approaches for heart sound recognition, the End-to-End approaches do not require all the steps in the pipeline including joint feature extraction, feature selection, and classification. It can implicitly extract strong generalized and abstracted features from raw data beginning from the low-level features until the

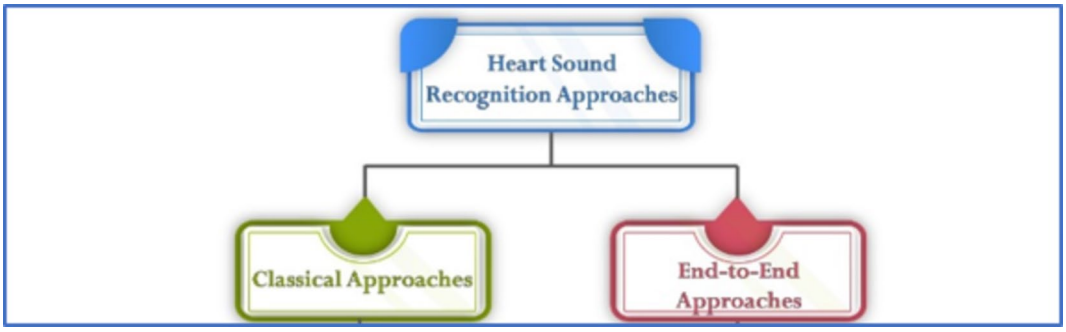

Fig. 1. Heart sound recognition approaches taxonomy.

high-level representation of the data instances. The simple End-to-End approach eliminates the need of complex feature engineering which is needed in classical approaches[7].

Classical approaches for heart sound recognition

Sharma et al.[8] adopted the Envelope-Based Segmentation method for segmenting heart sound signals. They applied the Hilbert energy envelope which was obtained by collecting the signal's instantaneous characteristics. The authors used two datasets, PASCAL, and a constructed database. By examining both datasets, the achieved accuracy was 94.34%. However, the proposed algorithm suffered from noisy peaks which affected distinguishing them from S1 and S2 peaks. To overcome the random noise problem of misidentification of S1 or S2, Plesinger et al.[9] made use of the ECG Reference-based Segmentation method. They applied FFT band-pass filter and Hilbert transformation to create five different amplitude envelopes with frequencies. The authors used the PhysioNet database. The naïve bayes classifier (NBC) was used for classification and the results show an accuracy of 0.841 (sensitivity 0.770, specificity 0.913). However, in point of view, combining the ECG with PCG for segmenting the heart sounds may increase the time complexity of the proposed method.

Oliveira et al.[10] presented a Probabilistic Model for FHS segmentation using a hidden semi-Markov model. The sojourn time distribution parameters were calculated and fine-tuned based on training and testing data using the Expectation-Maximization (EM) algorithm. The model consisted of four states, and they applied the Viterbi algorithm for hidden state sequence locating by searching for the most likely hidden sequence which maximizes the likelihood function. The authors used two publicly available datasets, PhysioNet and PASCAL, also they constructed a third paediatric dataset. The algorithm achieved an average F1 score of 92% however it suffered from increased computational complexity compared to other methods.

Studying Time-Frequency/Wavelet Analysis Based Methods has received attention in[11] who employed the DWT for the heart sound signal segmentation process. The authors extracted and selected features from the time domain and the frequency domain such as Shannon entropy and Spectral centroid. The system then used the extracted features for normal and murmur heart sound classification by applying Adaptive-neuro fuzzy inference system ANFIS and HMM with accuracies of 97.9 and 98.7% respectively. They studied three types of murmurs including mitral stenosis (MS), aortic regurgitation (AR), and mitral regurgitation (MR). The dataset was taken from 230 subjects with 150 normal heart sounds and 80 abnormal heart sounds.

Zeinali and Niaki[12] classified the heart sounds signals into three categories, normal, S3, and S4. They assessed three classification algorithms, SVM, gradient boosting algorithm (GBA), and random forest classifier. They applied wavelet features and Shannon information as Shannon entropy. Also, they extracted MFCC features to be compared to the previous features. The accuracy of the proposed methodology was found for binary classes, normal and abnormal with a score of 98% by applying the GBA using MFCC features. On the other hand, the GBA scored an accuracy of 87.5% using the proposed features and 95% using MFCC features. They used two datasets, PhysioNet and PASCAL. Wu et al.[13] proposed a methodology for classifying normal and abnormal heart sounds using complete empirical modal decomposition with adaptive noise (CEEMDAN) and SVM, and compared the SVM with the Fisher discriminant classification method. They decomposed the PCG signal into a few intrinsic mode functions (IMF), then they found the instantaneous frequency using Hilbert transform by optimizing the IMFs after applying the correlation coefficient, energy factor, and signal-to-noise ratio of the IMF and the original signal. They selected the 2016 PhysioNet/CinC dataset. The SVM scored better results than CEEMDAN-Fisher with an accuracy of 87%.

Khan et al.[14] presented a system for classifying normal heart sounds and Extrahls disorder. The classification methodology depended on providing six separate machine learning-based ensemble classifiers with extracted features. Different kernels have been used for K-nearest neighbor (KNN) classifiers including Boosted Trees (KNN-BoT), Bagged Trees (KNNBaT), Subspace Discriminant (KNN-SD), Subspace KNN (KNN-SK), and RVS Boosted Trees (KNN-RBT). The authors recorded PCG signals from 40 subjects, 20 normal and 20 had Exrahls. The methodology recorded cumulative accuracy of 98.8%, specificity of 97.56%, and sensitivity of 99.99%, with low processing costs caused by compressed features.

To distinguish between healthy PCG sounds and pathological sounds[15] developed an approach used to extract discriminative and stable features including Aortic Stenosis (AS), Mitral Regurgitation (MR), Mitral Stenosis (MS), and Mitral Valve Prolapse (MVP). Support Vector Machine (SVM) and KNN have been used with 64 features as input using ten-fold cross-validation. An open-access PCG sound dataset is issued by GitHub and used to evaluate the system which scored an accuracy of 99.5 and 98.30% using KNN and SVM classifiers, respectively. By utilizing the modified Empirical Wavelet Transform (mEWT) and the NASE, Narváez et al.[16] segmented the PCG signal to detect the systolic and diastolic intervals. They achieved a total error in the segmentation of 843,440.8 and 17,074.1 datasets A and B respectively for the PASCAL Challenge. For Normal and Abnormal classification, they selected samples from PhysioNet, PASCAL, the Database of the University of Michigan, the Database of the University of Washington, the Thinklabs database, and the 3 M database. The results from the KNN classifier showed that the accuracy is 99.26%, a specificity is 100%, and a sensitivity is 98.57%.

Bao et al.[17] used different deep learning approaches including CNN, and RNN models (LSTM, BiLSTM, GRU, and BiGRU) in order to investigate the heart sound signal duration. To classify the heart sound durations into normal abnormal PhysioNet dataset has been used. After that, they compared between the performance of the approaches used and found the best performance was obtained by the GRU with an accuracy of 94.07%, a specificity of 94.81%, a sensitivity of 91.29%, and an overall score (MAcc) of 93.05%.

To classify five categories of PCG signals, healthy, aortic stenosis, mitral stenosis, mitral regurgitation, and mitral valve prolapse, Arslan and Karhan[18] used the GitHub public dataset for training and testing. They used Hilbert-Huang transform to decompose each signal into its intrinsic mode functions (IMFs). For each decomposed mode, they extracted the MFCC features and selected the representative features using the Genetic

algorithm. In the classification step, they compared four classifiers, the KNN, the multilayer perceptron (MLP), the SVM, and the DNN. The system scored the highest results using the DNN with a precision of 98.9%, recall of 98.7%, F1 score of 98.8%, and accuracy of 98.9%.

End-to-end approaches for heart sound recognition

Shuvo et al.[19] used the raw PCG signal as input to their system to overcome hand-engineering feature extraction techniques that involve computational complexity and time-consuming issues. The system was used to classify five categories of heart sounds, including N, AS, MR, MS, and MVP. They applied two steps for learning and feature extraction using three parallel CNNs, two of them were 1D-CNN and the third one was 2D-CNN, and one LSTM. The outputs from the three parallel CNNs were combined to enter an input to the LSTM. They used PhysioNet and GitHub datasets and achieved an accuracy of 88.09%, precision of 88.08%, recall of 87.98%, and F1 score of 88.03%.

Another raw PCG input method for classifying normal and pathological heart sounds without segmentation was adopted by Mehmet Bilal[20]. They used two 1D-CNNs for feature extraction, the first 1D-CNN focused on the texture-based approach to extract One Dimensional local binary pattern (1D-LBP) and the other for capturing local ternary patterns (LTP). The author used PASCAL and PhysioNet for training, validation, and testing operations. The system acquired accuracy rates of 91.66 and 91.78% for PASCAL and PhysioNet datasets respectively.

Abnormal heart sound recognition methodology

Figure 2 represents the overall research design, which represents the general architecture of the proposed abnormal heart sound recognition framework.

This research study consists of two main phases which are organized as shown in (Fig. 3).

Pre-processing and models training

The requirement of having instantaneous recognition of Abnormal heart sounds is based on developing a real-time system with low computational cost. This can be achieved by applying classical machine learning algorithms, specifically the SVM classifier. In abnormal cases, there is a need for determining the type of abnormality which can take more time than detecting normal or abnormal cases. For this purpose, we apply the LSTM classifier to determine the abnormality class.

SVM training

The PhysioNet dataset contains both Normal and Abnormal cases, which is used as a training dataset for our model. The stages are demonstrated in (Fig. 4).

Sampling and filtering The input analog heart sound signal (x(t)) is first converted into a digital signal (X(n)) using an analog-to-digital converter (ADC). The digitization begins with sampling which converts the continuous-time signal into a discrete-time signal of 4 kHz sampling rate or sampling frequency (fs) then quantization which converts the continuous amplitude into a discrete with 16 bits of resolution[21,22]. Since signals from different datasets are sampled at varying frequency rates, it is necessary to down-sample each signal to 4 kHz.

The PCG signal may contain noises or artifacts due to instrument noise, noise from body sounds such as respiratory noise, environmental background noise, noise due to subject motion, and movement of stethoscope diaphragms. We applied two second-order Butterworth bandpass filters Infinite Impulse Response (IIR) Filters due to their less computational time[23]. The Butterworth filters are used for noise and artifact removal from the

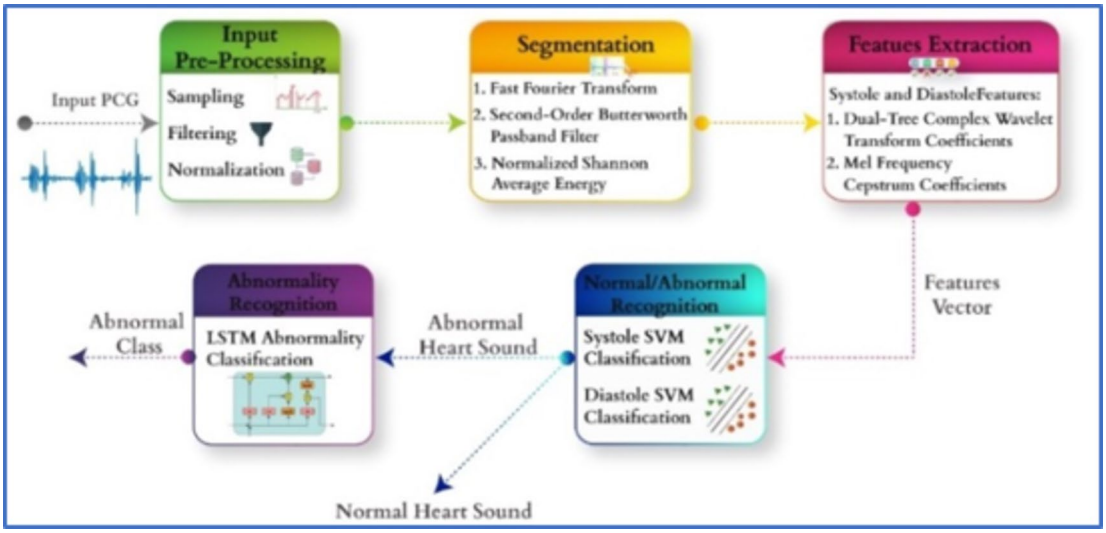

Fig. 2. Abnormal heart sound recognition framework.

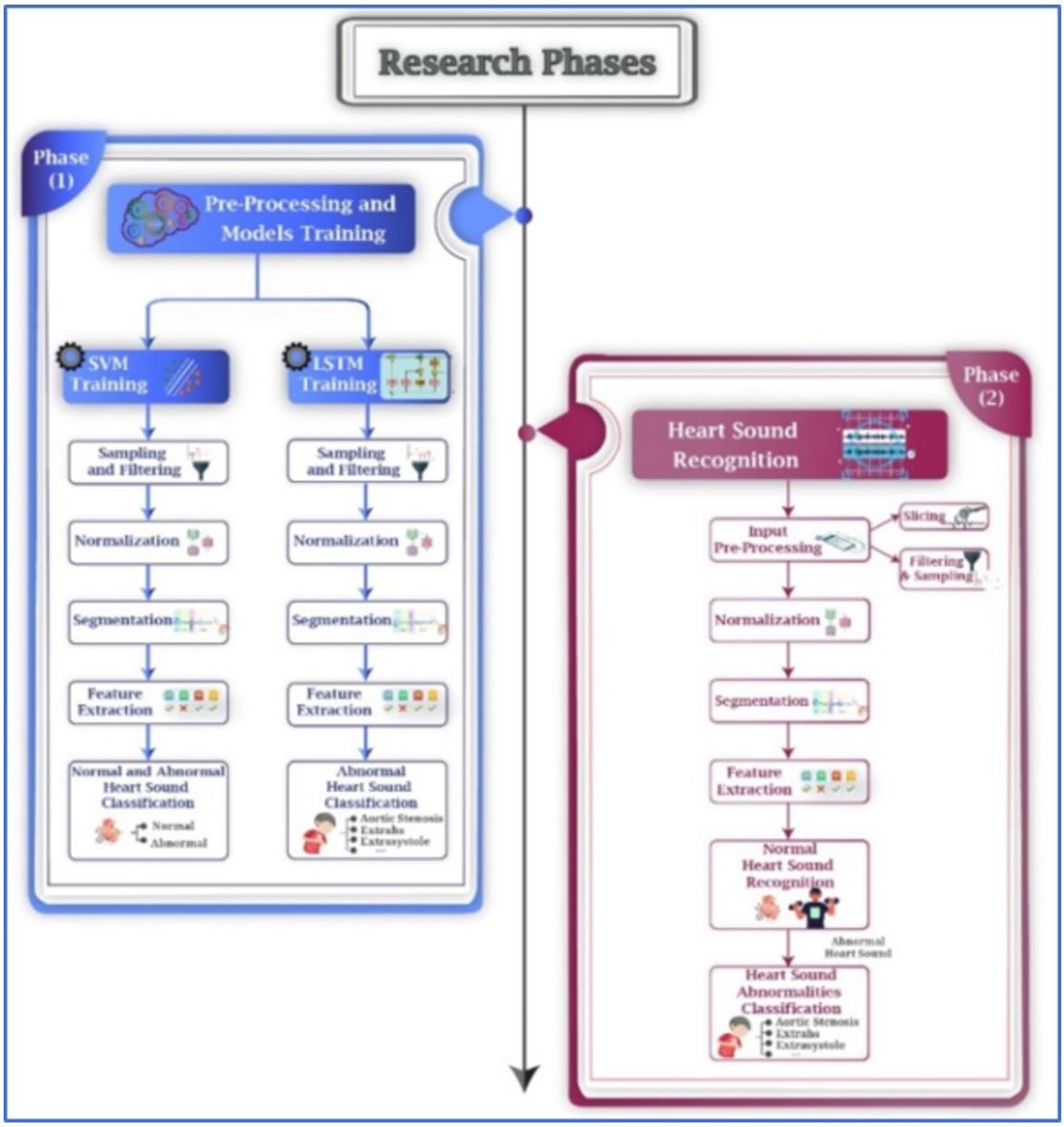

Fig. 3. Research phases for heart sound recognition system.

input digital PCG, and they are distinguished by their maximally flat passband with 0% ripples to enable the filtered part of the signal to pass unaltered. The first filter has a low cut-off frequency (fL) of 20 Hz and a high cut-off frequency (fH) of 200 Hz, and the output is used for the segmentation process to decrease the computational time by obtaining the frequency range for S1 and S2. While, the second filter has an fL of 20 Hz and an fH of 1000 Hz[24,25], which is in the range of normal and abnormal heart frequencies and eliminates noise characterized by high frequencies.

Normalization Normalization eliminates the amplitude variation from subject to subject and enhances classification. We transfer the peak amplitude to a reference level of [-1,1], this is done by dividing each PCG sample by the maximum value of the PCG sample in each heart sound, as in Eq. (1). Where X(n) is the filtered PCG sample and XNorm(n) is the normalized one.

$$X_{Norm}(n) = \frac{X(n)}{Max_{n=1}^{N} |X(n)|} \tag{1}$$

Segmentation PCG segmentation is critical in controlling the heart sound feature extraction and classification by dividing the signal into its fundamental components the S1, S2, systole, and diastole with statistical character-

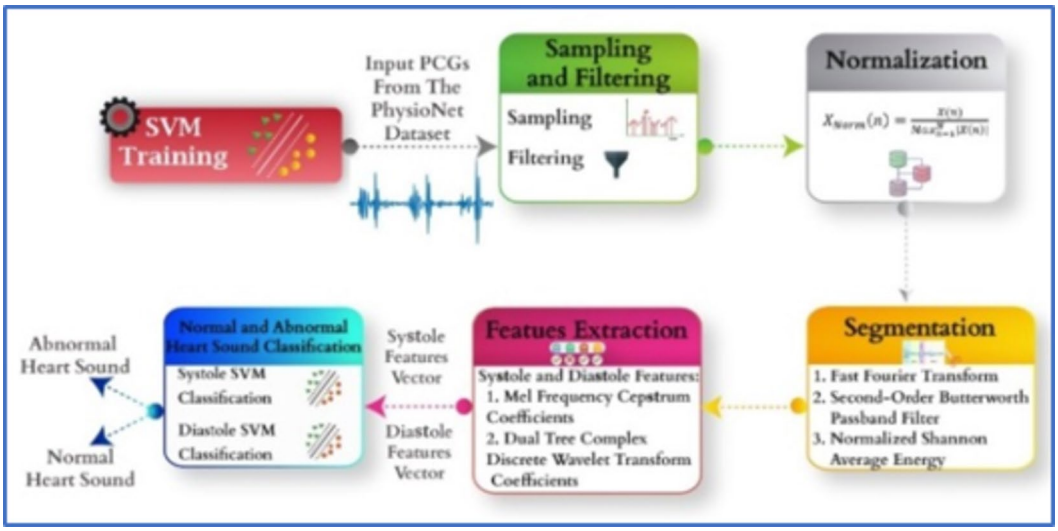

Fig. 4. SVM training stages.

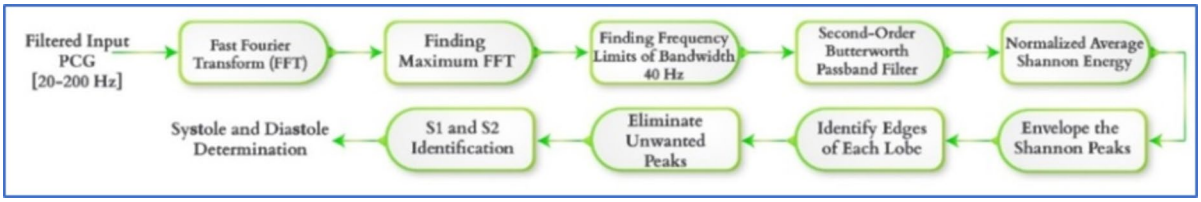

Fig. 5. Steps of FFT and the normalized average shannon energy segmentation process.

izations for each component. Recent studies showed the effectiveness of two approaches, EMD and EWT, both of them are adaptive but EMD is not theoretically supported, and it has a high computational cost compared with EWT. EMD is difficult to understand or obtain relevant information from the signal due to decomposing the signal in many ways. Also, for real-time systems EWT still has a moderate computational cost, to improve this method, a modified version of EWT is released (mEWT) which can specify the boundaries and the edge selection method. In our work, we got the benefits of the idea of mEWT and applied another second-order Butterworth passband filter, which depended on FFT to determine the low and the high-frequency range for the Butterworth filter. This new method decreases the computational time for the segmentation process. We also applied the NASE with integration as shown in (Fig. 5). According to a critical review of PCG segmentation algorithms, the Shannon Energy Envelogram scored the highest accuracy of 99.2%[26].

The PCG signal is composed of N mono-components and each of these components is composed of N segments based on Fourier Spectrum. The spectrum is obtained by the FFT, then computing the local maxima (the center frequency of the filter) in the spectrum which is used to build boundaries by defining all the local minima in the global trend. The boundaries for the Butterworth passband filter in the segmentation process depend on the output of the FFT, to cover the variations between individuals and to extract representable components from the PCG signals. After finding the spectrum by the FFT, the maximum values are to be as center frequencies for the Butterworth filter. The filter bandwidth of 40 Hz is applied which provided good coverage and representation for S1 and S2[16]. For instance, if the local maximum is 80 Hz, then the range for the filter is from (0–40/2,80 + 40/2) which gives us (60–100) and can reduce the computational cost by having an optimal filter that contains both S1 and S2 with the clearest form.

After obtaining the fundamental component including S1 and S2, we need to find an envelope to highlight S1 and S2 peaks and remove any other non-related peaks, either based on the energy values or based on the time interval. Before we segment the desired component, we should build up an envelope surrounding S1 and S2. The most efficient way is by applying the Shannon energy to the selected component of the PCG signal. Shannon energy can represent and recognize the magnitude of oscillations of high intensity as well as those in low amplitudes. The Shannon energy is defined by Eq. (2).

$$\boldsymbol{E} = -\boldsymbol{x}\left(\boldsymbol{i}\right)^{2}\boldsymbol{log}\left(\boldsymbol{x}\left(\boldsymbol{i}\right)^{2}\right) \tag{2}$$

where $x\left(i\right)$ represents the samples of the signal and E is Shannon's energy.

The squaring operation in Shannon's equation is able to enlarge the values and may make low amplitude S1 or S2 to be faint and may not appear. In order to solve this problem, the NASE is used. First, we calculate the

average Shannon energy using Eq. (3) by dividing the signal into frames, with each frame duration of 0.02 s and overlapping 0.01 s[27,28].

$$Es = -\frac{1}{Ms}\sum_{i=1}^{Ms} {x_{norm}}^2(i) \,.\, log{x_{norm}}^2(i) \tag{3}$$

The second step includes applying the normalization equation which depends on calculating the mean of energy values, and the standard deviation of the energy frames, as in Eq. (4):

$$En(t) = \frac{Es(t) - \mu(t)}{\sigma(t)} \tag{4}$$

where $\mu(t)$ is the average value of energy Es of the signal, $\sigma(t)$ is the standard deviation of energy Es(t) of the signal and En(t) is the NASE.

After calculating NASE, we applied integration by convolving the signal with ones to obtain the boundary of each heart sound accurately, then a zero-crossing algorithm is used to know the starting and stopping points of each heart sound by calculating the points where the sign of the boundary's changes from positive to zero or vice versa. After identifying the edges of each lobe, some of the lobes are eliminated if they have short durations and low energy. On average, the normal cardiac cycle lasts for 0.8 s, with a systolic interval of 0.2 s approximately, and a diastolic period of 0.6. If the time interval between two peaks is less than 0.2, it is eliminated.

After studying the files in the PhysioNet dataset, we applied a minimum thresholding value for the calculated NASE to eliminate any peaks having energy less than 0.07. By establishing another condition to solve the problem of low amplitudes of S2, if the interval is more than 0.65 s, then there exists a peak for S2. To differentiate between S1 and S2 peaks in the PCG signal, the assumption of this study is that the systolic interval is shorter than the diastolic interval. The Systolic interval occurs between S1 and S2 while the diastolic one occurs between S2 and S1 of the next cycle.

Feature extraction The PCG is a stochastic non-stationary signal, which cannot be written in a mathematical equation and its statistical properties change over time. For this reason, we need to find a representation for the PCG signal by having an appropriate windowing procedure to divide this non-stationary signal into segments or parts that have stationary properties. There are different domains for exploring and efficiently capturing various physiological and pathological characteristics and features. The time domain has simple extracting features, and the frequency domain has some useful information about patterns in the signal, to combine the powerful representation of both domains the time-frequency domain has been reported to have better feature representation results.

By reviewing the literature, the most common and efficient ways for feature extraction are the MFCCs and the DWT coefficients, which can capture the non-stationary properties of the PCG signal and represent information in the time-frequency domains efficiently. The DWT suffers from four main problems, oscillations, shift variance (small shift of the signal can disturb the wavelet coefficients), aliasing, and lack of directionality[16].

To solve the above-mentioned issues of DWT, we have applied the DTCWT to our work, which is approximately shift-invariant and has anti-aliasing properties. We have combined the DTCWT with MFCC to have a representative feature vector, by taking the MFCC coefficients for the three-level DTCWT. After the segmentation process, we adjust the regions of the second filtered and normalized input PCG, which has a frequency range of 20–1000 Hz relying on the first segmented PCG, to extract features for systole and diastole segments.

- Dual-tree complex wavelet transform (DTCWT) coefficients

Biological and medical signals such as PCG and ECG are non-stationary signals, therefore they require types of analysis considering their changes through time and dividing the non-stationary signals into segments with stationary properties. Fourier transform (FT) can represent only global features in the frequency domain regardless of time, while the STFT takes into consideration the frequency characteristics related to time changes by windowing the signal and applying FT on each window. STFT has a fixed window size which is considered a limitation. The wavelet transform (WT) appeared as an improved version of STFT by having various window sizes and locations. WT is used broadly in several medical applications for its ability of multiresolution analysis and different resolutions at different frequencies. It is considered one of the best transforms for analysing transient and nonstationary signals since it offers a sufficient resolution in both the time and frequency domains. It uses narrower windows at high frequencies for better time resolution and wider windows at low frequencies for better frequency resolution[29].

WT can extract local spectral and temporal information simultaneously by scaling and translating the wavelets on the desired signal with a variety of mother wavelet options to be chosen based on the input signal characteristics, such as Daubechies, Morlet, and Haar. There are two versions of WT, the continuous WT (CWT) which suffers from redundancy and is considered to have higher computational times, and the discrete WT (DWT) which can be convenient for real-time systems, however, it suffers from oscillations, shift variance, aliasing, and lack of directionality. To overcome these problems, in 1998, Kingsbury introduced DTCWT[30].

DTCWT is proper for signal and image processing applications, which was first applied in image processing. It utilizes two DWTs with specific settings to form approximately a Hilbert transform pair. DTCWT divides the input signal into two trees, the real and imaginary trees by applying dual-tree of wavelet filters as shown in (Fig. 6). The DTCWT approximately considered shift in-variance, anti-aliasing, and directionally selective filters.

In each tree, the DTCWT applies filter banks on each input to produce high-frequency and low-frequency sub-bands containing half of the samples (downsampled by 2) with different cut-off frequencies and scaling. The compression of the applied wavelet represents high-frequency bands, while dilation produces low-frequency bands. This procedure is known as one-level decomposition and can be expressed by Eqs. 5 and 6.

$$y_{high}[k] = cA_{\psi(n)} = \sum {}_n x[n] * \psi[2k-n] \tag{5}$$

$$y_{low}[k] = cD_{\phi(n)} = \sum {}_n x[n] * \phi[2k-n] \tag{6}$$

where $x[n]$ is the input to the specified level, $y_{high}[k]$ is the output of the high pass filter and it is called detail coefficients (low-scale, high-frequency components), $\psi[2k-n]$ is the transfer function for the high pass filter, $y_{low}[k]$ is the output of the low pass filter and it is called approximation coefficients (high-scale, low-frequency components of the signal), and $\phi[2k-n]$ is the transfer function for the low-pass filter[31].

In the proposed work, the signal is decomposed into three-level, which provided the best representative decomposition and clarity for the signal. We take the real tree coefficients of level 3 including both the approximation and detailed coefficients. Those utilized coefficients are used as an input to the MFCC process to extract meaningful features.

- Mel frequency cepstral coefficients (MFCCs)

MFCCs have many applications in speech processing and emotion recognition due to their effectiveness in sound and audio data. This technique showed promising results in PCG signal by reflecting the nonlinear relationship between the human ear and the frequency of the heard sound. The steps for MFCC extraction are illustrated and shown in (Fig. 7).

For comparison purposes, we are using several feature vectors either for one cardiac cycle or for three cardiac cycles as shown in (Table 1). There are two feature vectors obtained from one cardiac cycle segmented into a systolic and a diastolic period with a length of 14 MFCCs for the systolic and 33 MFCCs for the diastolic. Another option has also two feature vectors for three cardiac cycles segmented into systolic and diastolic had a

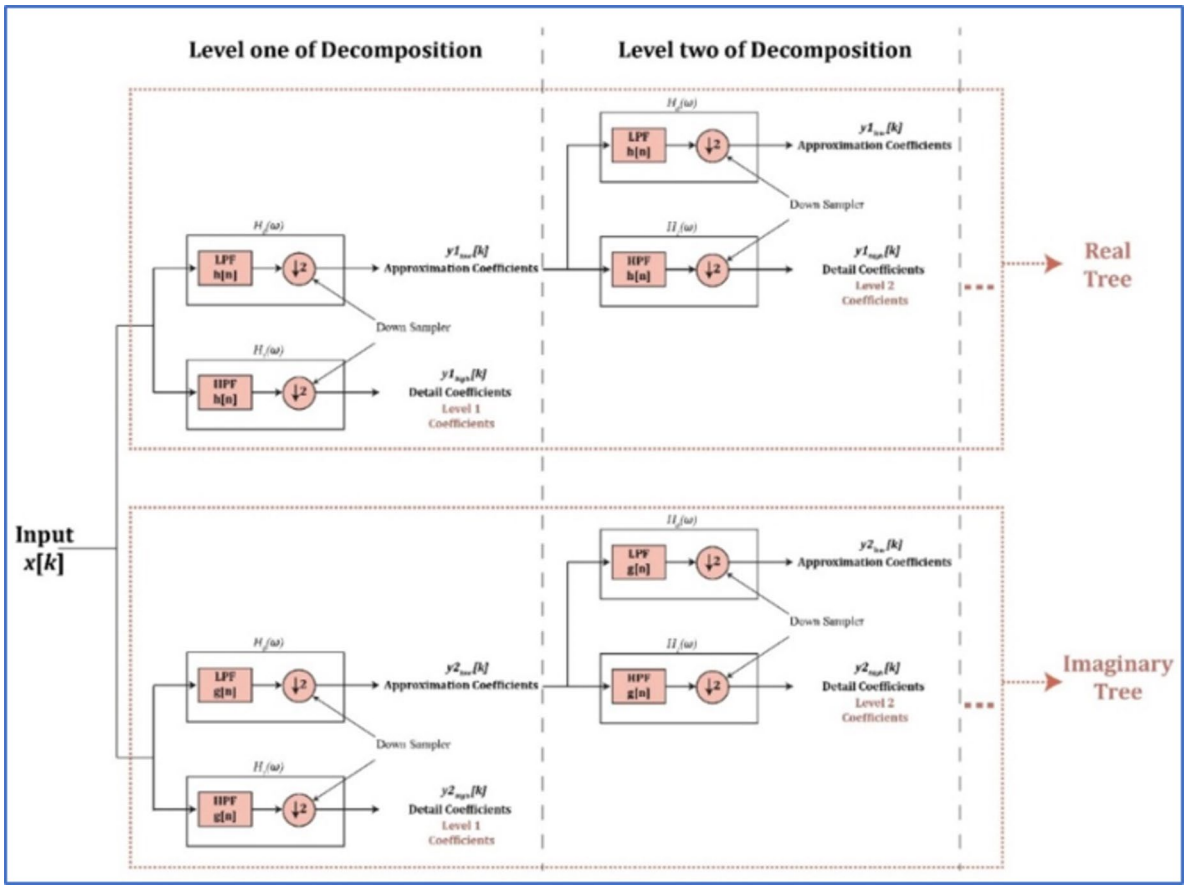

Fig. 6. Filter banks with high pass and lowpass sub-bands for real and imaginary trees of DTCWT coefficient's extraction.

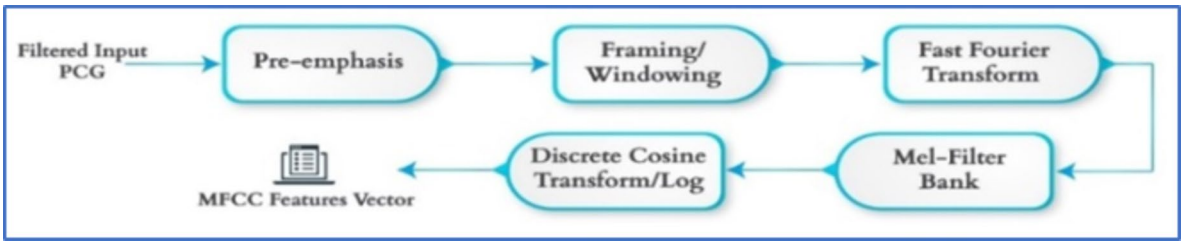

Fig. 7. MFCC feature extraction method.

	Sampling freq. for MFCC	Length of feature vector
Single systolic feature vector	2000	14 × 14
Single diastolic feature vector	4000	33 × 14
Three systolic feature vector	10,000	51 × 14
Three diastolic feature vector	10,000	71 × 14
Three systolic + three diastolic feature vector	10,000	42 × 14

Table 1. Summary of feature vectors.

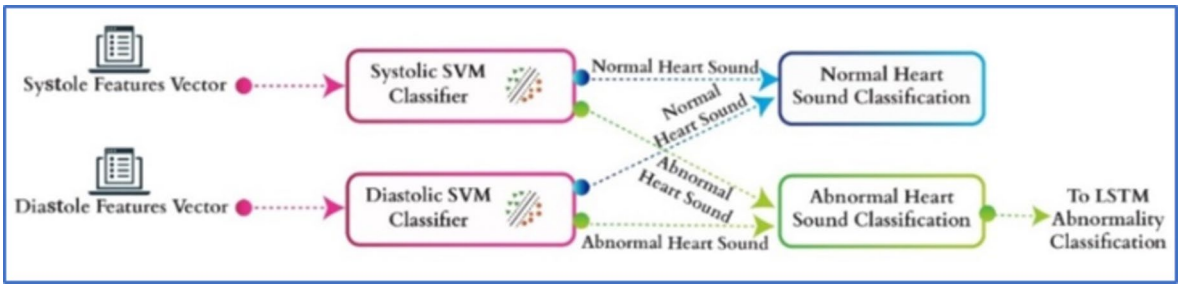

Fig. 8. Normal and abnormal heart sound classification.

length of 51 MFCCs for the systolic and 71 for the diastolic. The final case uses three cardiac cycles segmented into systolic and diastolic combined in one feature vector of 42 MFCCs.

Normal and abnormal heart sound classification The proposed algorithm classifies PCG signals into normal or abnormal sounds in a real-time environment. To achieve this, two SVMs are employed. Each SVM specializes in analyzing feature vectors derived from either the systolic or diastolic periods of the cardiac cycle. The two SVMs will be trained on the two feature vectors for the systolic and diastolic periods. Those vectors are the result of combining the MFCC and DWT extracted coefficients. This dual-SVM methodology permits the separate assessment of systolic and diastolic characteristics, ensuring precise classification for each phase of the cardiac cycle. Also, experiments are made to compare two feature vectors using two SVMs with using a single feature vector using a single SVM.

SVM is a supervised machine learning algorithm that was first reported by Cortes and Vapnik[32] for binary classification problems, then it was extended to multi-class classification problems. According to the collected literature, SVM is suitable for classifying Normal and Abnormal heart sounds even if we have high dimensionality classification problems and small sample sizes. It is a linear classifier and a non-linear classifier (using kernels) that builds a model by finding a separator hyperplane in a higher dimension. The dimension of the hyperplane is determined by the number of extracted features. Dividing the features vector and using two SVMs can reduce the computational time and find better prediction results.

The PhysioNet dataset is used for training the two SVMs, and if the output of one of them is Abnormal then the system alerts the patient of having a problem with the PCG signal and needs help. The system enters the next step of determining the type of abnormality using LSTM, but if the PCG signal is Normal, then there is no need for any further steps, as shown in (Fig. 8).

LSTM training

The PASCAL and GitHub datasets contain six classes of Abnormal heart sounds for various types of Murmurs, Extrahs (extra heart sound signal), and Extrasystole (a heart sound that is out of rhythm involving extra or skipped heartbeats)[33] which is used as a training dataset for our model. The first four steps in the SVM Training

phase are applied to the two datasets including, sampling and filtering, normalization, segmentation, and feature extraction as shown in (Fig. 9). The features vector includes the MFCCs of three-level DTCWT coefficients combining both systole and diastole characteristics for distinguishing between the six Abnormal PCG categories.

Heart sound signals are a kind of sequential data with a strong temporal correlation. RNNs are capable of handling temporal data and providing better predictions by learning the temporal context of input sequences, but they suffer from the vanishing gradient problem which happens due to many operations which end up with a vanishing value. To solve this problem, LSTMs were introduced by Hochreiter and Schmidhuber[34] who added memory cells that can maintain information in memory for long periods. LSTMs are capable of selecting what information to be retained and what unhelpful information to be removed using forget gates. LSTM consists of three main cell-memory networks and three gates, the input gate, the forget gate, and the output gate[35].

Finally, the LSTM classifies the six Abnormal classes after being trained on the two datasets. Applying a deep learning algorithm can cause high computational time compared to the classical machine learning algorithm, however, recognition of the abnormality type is not as urgent as detecting abnormal cases because the patient is already on his/her way to getting help.

Heart sound recognition

The heart sound detection phase aims to classify any input PCG signal by first pre-processing, segmenting, and feature extracting of the input to enter the Normal heart sound recognition and if the output was an abnormal class then this PCG enters abnormality classification to determine the type of abnormality.

Input pre-processing

The input PCG signal is passed through a series of steps by slicing the input, filtering it from noise, and sampling it to be prepared as an input for the next stages.

Slicing Each cardiac cycle is approximately 0.8 s on average[35]. According to the cardiac cycle duration, the input PCG signal is sliced into segments of 0.8 s to capture the features of the events from the beginning of one heart contraction to the beginning of the next one.

Filtering and sampling The sliced input PCG signal is filtered as described before in the Sampling and filtering section.

Normalization

The filtered PCG signal values are normalized as mentioned in the pre-processing and model training Normalization subsection.

Segmentation

The filtered PCG is segmented based on the FFT and filtering methods into four segments, S1, systole, S2, and diastole. The details for this process are described in the pre-processing and models training Segmentation subsection.

Feature extraction

The segmented input PCG enters the feature extraction step by applying the MFCC to the DTCWT to have two feature vectors, one for the systolic and the other for the diastolic interval, as input for the classification steps. For more details please refer to the pre-processing and models training Feature Extraction subsection.

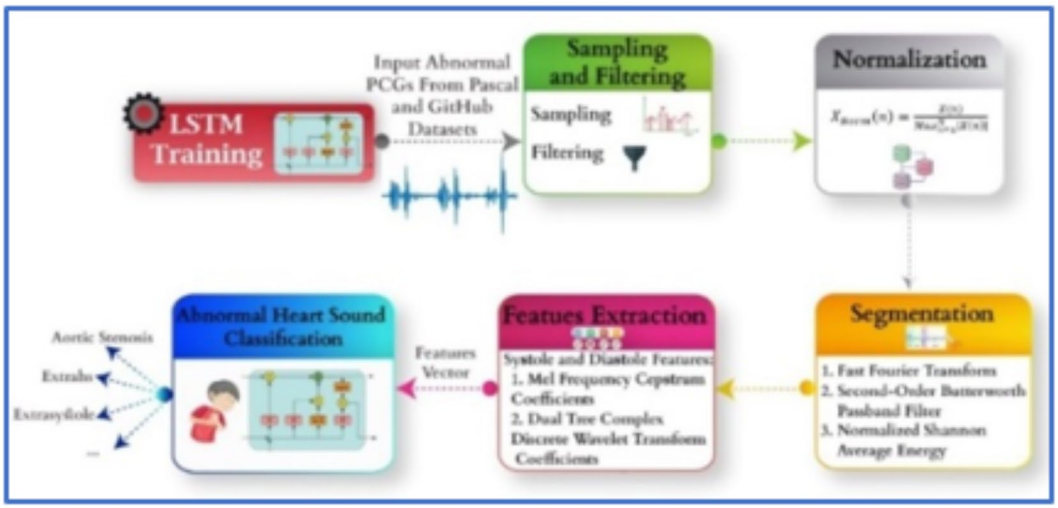

Fig. 9. LSTM training stages.

Normal heart sound recognition
The two feature vectors are an input for the first and second SVMs to distinguish between normal and abnormal sounds. If the output of one of those SVMs gave an abnormal class, then the final result forces the system to enter the heart sound abnormality classification to determine the class of this abnormality.

Heart sound abnormalities classification
In case of an abnormal result, the two feature vectors are combined to be an input for the LSTM classifier to detect the abnormality type, either one of the six mentioned classes (AS, MR, MS, MVP, Extrahls, and Extrasystole).

Datasets
This research paper uses three different datasets to train, validate and evaluate the proposed system. The PhysioNet[36] dataset is used for the Normal/Abnormal recognition phase, while PASCAL[37] and Open-Access GitHub[38] datasets are used for Abnormality classification.

The training and validation dataset is divided into two datasets, the first dataset is used for Normal and Abnormal training using the PhysioNet dataset with 1116 files, while the second dataset is used for determining abnormality categories including the four murmur types (AS, MR, MS, MVP) using the GitHub dataset with 640 files, and the other two types of abnormalities (Extrahls and Extrasystole). To train the models for abnormality categories, there are two ways due to the unbalanced number of files in the abnormal databases. To detect the four murmur abnormalities, the number of files for each category is 160 files. While to detect the overall six types of abnormalities, the number of files for each category is 40 files selected totally from the PASCAL dataset and randomly from the GitHub dataset.

The testing dataset is divided like the training dataset with 279 files from the PhysioNet, 160 files from the GitHub dataset, and 20 files from the PASCAL dataset. There are experiments for the four murmur abnormalities and the overall six types of abnormalities. In the first experiment, the number of files for each category is 40 files, while in the second experiment, the number of files for each category is 10 files.

The PhysioNet dataset contains 3122 heart sound recordings distributed in five sub-folders labeled from A through E, with variable lengths ranging from five seconds to 120 s divided into 2,495 Normal PCGs and 627 Abnormal PCGs[39]. They were recorded in clinical or non-clinical environments among patients and healthy participants including children and adults. It was introduced in 2016 for PhysioNet/CinC Challenge, and it was auscultated and recorded from four locations on the body, namely the aortic area, pulmonic area, tricuspid area, and mitral area. Each audio recording includes only one PCG signal with a sampling rate of 2000 Hz in .wav format.

The PASCAL database is composed of 832 recordings divided into two datasets, A and B with .wav and .aif formats[40]. Dataset A contains 176 files distributed into four categories, Normal (31 files), Murmur (34 files), extra heart sound (Extrahs with 19 files), Artifact (40 files), and unlabelled (52 files). On the other hand, Dataset B includes 656 files categorized into four groups, Normal (320 files), Murmur (95 files), Extrasystole (46 files), and Unlabelled (195 files). The recordings have a time length ranging from 1 s to 30 s with a frequency range below 195 Hz.

The Open-Access GitHub dataset contains five classes, Normal, Aortic Stenosis, Mitral Stenosis, Mitral Regurgitation, and Mitral Valve Prolapse[41]. They collected 1000 audio files in .wav format, and they divided the recordings into five equal categories (200 files for each class). They collected data from random sources, such as books and websites and it was sampled to 8000 Hz.

Each file in the Open-Access GitHub dataset is divided into three cardiac cycles (three of S1 and three of S2), with an average length of 2.4 s. By applying the segmentation method, we have segmented the input files from the two datasets, the PASCAL and the PhysioNet, into three cardiac cycles consistent with the GitHub dataset. The final dataset consisted of 1400 files from the PhysioNet divided equally into Normal and Abnormal, and 100 files from the PASCAL dataset divided equally into Extrahls and Extrasystole.

Results
Experiments and tests have been conducted for both segmentation and classification modules to estimate the validation of the entire system. We started by testing the segmentation method based on two datasets, the PhysioNet and PASCAL by comparing the outputs from our module and their segmentation outputs by observing the performance accuracy for the module itself and after integrating it with the classification modules. The classification accuracy and other measures were evaluated for the Normal/Abnormal classes and the Abnormality type recognition.

Segmentation results
To test the performance of the segmentation method that is responsible for locating S1 and S2, we used two datasets the PASCAL and the PhysioNet datasets. They were used to obtain files of three cardiac cycles. The performance was assessed using two different ways, for the PASCAL we used Eqs. 15 and 16 which were applied in the PASCAL challenge (which provided the reference location of S1 and S2 in ms), also we used the ECG as a guide for the PhysioNet dataset. In Fig. 10, an example of abnormal heart sound segmentation was obtained from the PhysioNet dataset.

$$\text{Average Error for single PCG} = \frac{\sum \boldsymbol{abs\,(calculated\ location\ of\ S1\ or\ S2 - reference\ location\ of\ S1\ or\ S2\,)}}{\boldsymbol{Total\ Number\ of\ S1\ and\ S2}} \tag{7}$$

$$\text{Total Error} = \sum \boldsymbol{Average\ Error\ for\ all\ PCGs\ in\ datasets\ A\ and\ B} \tag{8}$$

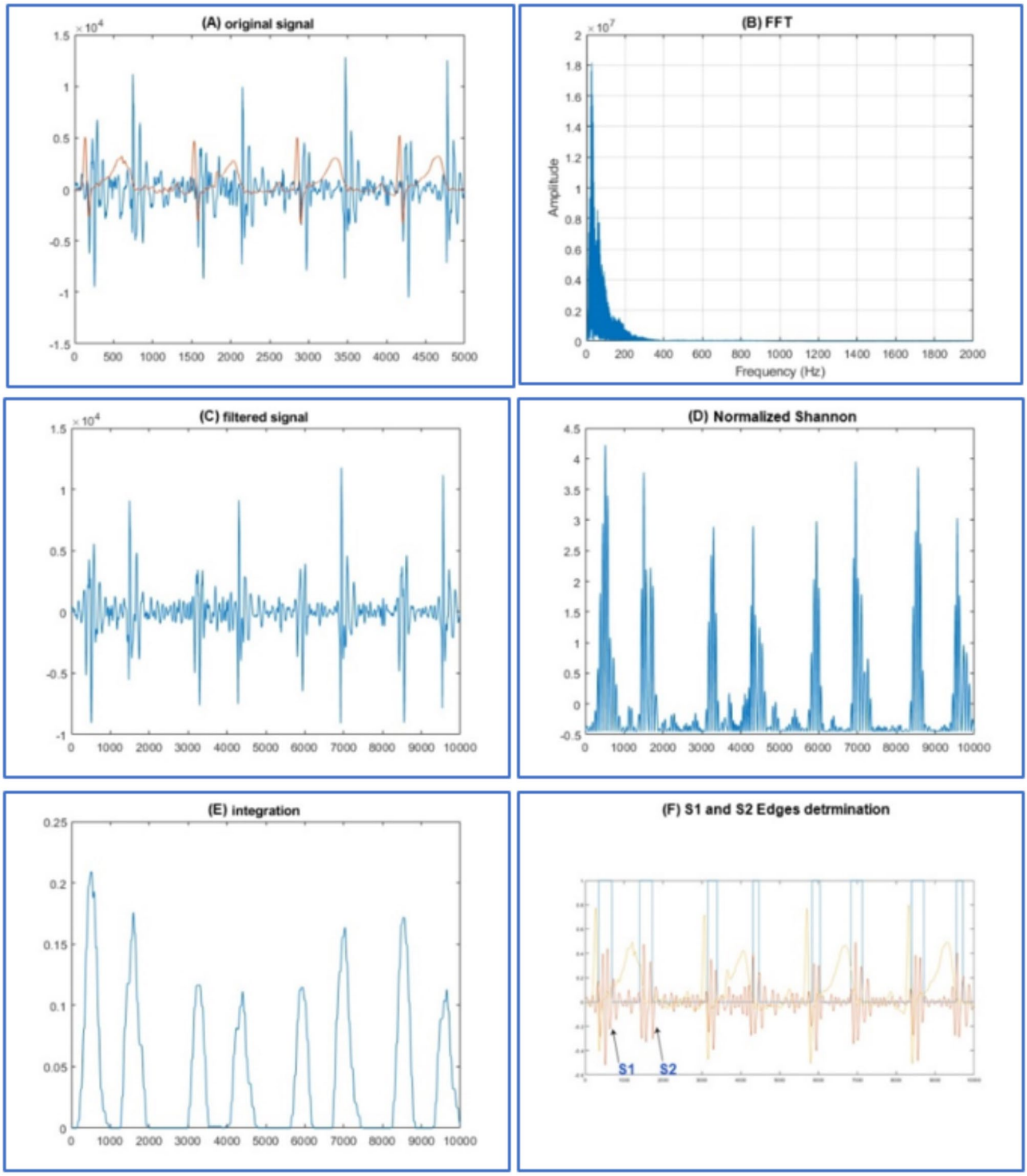

Fig. 10. Segmentation stages for a PCG signal (**A**) Normal heart sound recording with its corresponding ECG (**B**) FFT of heart sound (**C**) Filtering signal with a bandwidth of 40 Hz (**D**) Applying NASE for the filtered signal (**E**) Integration to envelope the peaks (**F**) S1 and S2 edges determination.

As is observed in Table 2, the segmentation results for four files are obtained from PASCAL datasets A and B. As it is stated in[16], the results of the total errors shown in[16] and the proposed method for datasets A and B are shown. Considering that the results presented in[16] were the best obtained in this dataset. Our method obtained a total error of 867,525.6 ms for dataset A and 23,590.3 ms for dataset B which have approximately similar results to that[16] and better than results presented in[42] as described in (Table 3). The differences did not affect on the extracted features and the overall determination of S1 and S2. By calculating the computational time required for segmenting files using the mEWT and the proposed approach, we have found that to segment a file of 30.79 s, the

Dataset	HS (file name)	HB (HeartBeat)	Average error (ms)
A	201,101,070,538.wav	11.5	19176.7
A	201,101,151,127.wav	4	917.6
B	103_1305031931979_B. wav	12.5	48.2
B	103_1305031931979_D2.wav	10	35.3

Table 2. Results of segmentation for samples from dataset A and dataset B.

Method	Dataset A (ms)	Dataset B (ms)
[16]	843,440.8	17,074.1
[42]	873,577.9	29,269.6
The proposed approach	867,525.6	23,590.3

Table 3. Total error for datasets A and B.

Dataset	PhysioNet		GitHub		PhysioNet and GitHub	
Classifier type	LSTM (%)	SVM (%)	LSTM (%)	SVM (%)	LSTM (%)	SVM (%)
Testing accuracy	90.0	90.7	99.6	99.6	92.2	90.7
Overall accuracy	97.3	98.1	99.6	99.9	98.0	98.1
Testing sensitivity	91.5	91.5	100.0	100.0	93.4	84.2
Testing specificity	88.4	89.9	99.3	99.3	90.6	98.4

Table 4. Classification results for normal and abnormal classes from the PhysioNet dataset using a single feature vector (one SVM).

	One systolic		One diastolic		IF output	
	LSTM (%)	SVM (%)	LSTM (%)	SVM (%)	LSTM (%)	SVM (%)
Testing accuracy	88.8	82.2	82.6	78.3	86.1	80.2
Overall accuracy	97.7	96.4	96.4	95.7	97.2	96.0
Testing Sensitivity	81.7	83.8	81.7	81.5	96.7	94.4
Testing specificity	85.1	80.5	83.5	74.8	74.8	64.9

Table 5. Classification results for normal and abnormal classes from the PhysioNet dataset using a single cardiac cycle with two feature vectors (two SVMs).

mEWT required 14.5 s to segment the whole file into 30 cardiac cycles, while the proposed approach required 1.2 s. On average, the consumed time for segmenting one cardiac cycle is 0.49 s for mEWT and 0.04 for the proposed approach.

For performance comparison based on computational time and classification measures, we utilize the segmentation, features extraction, and classification methods on a single combined feature vector consisting of systolic and diastolic segments to use them as input for either a single LSTM or SVM as demonstrated in (Table 4). For the PhysioNet dataset, the testing accuracy, sensitivity, and specificity scores are 90.0, 91.5, and 88.4% for a single LSTM, while the scores for a single SVM for the same dataset are 90.7, 91.5, and 89.9%. The scores obtained from the GitHub dataset are the same for a single LSTM or a single SVM which are 99.6, 100.0, and 99.3% respectively. The final results are for combining the PhysioNet and the GitHub randomly selected files, which provide accuracy, sensitivity, and specificity for a single LSTM of 92.2, 93.4, and 90.6%, while for a single SVM the scores are 90.7, 84.1, and 98.4% correspondingly.

As it is observed in Table 5, testing a single-segmented cardiac cycle separated into two feature vectors, one for the systolic and the other for the diastolic, achieves a final accuracy of 86.1% for the LSTM classifier, and 80.2% for the SVM classifier. The sensitivity of the LSTM and the SVM classifiers are 96.7 and 94.4% respectively. While the specificity scores for the LSTM and the SVM classifiers are 74.8 and 64.9% respectively.

Figures 11 and 12, and Fig. 13 summarized the overall accuracy and testing sensitivity for Normal/Abnormal PCG classification based on a single cardiac cycle and two feature vectors, on three cardiac cycles and two feature vectors, and a single cardiac cycle and single feature vector respectively.

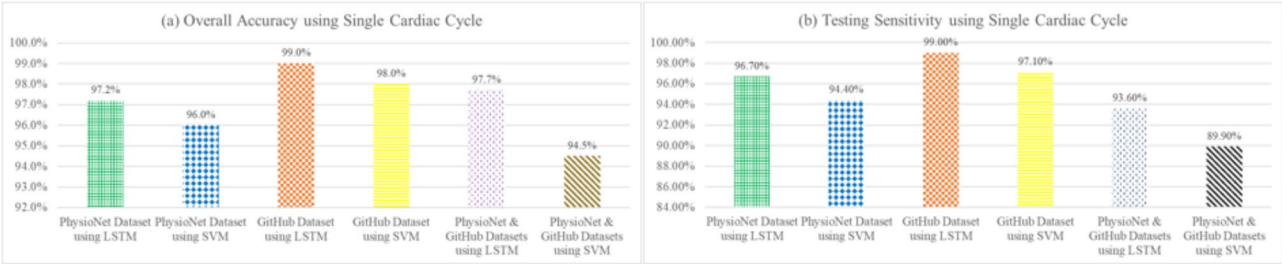

Fig. 11. (**a**) Overall accuracy (**b**) testing sensitivity for normal/abnormal PCG classification based on a single cardiac cycle and two feature vectors.

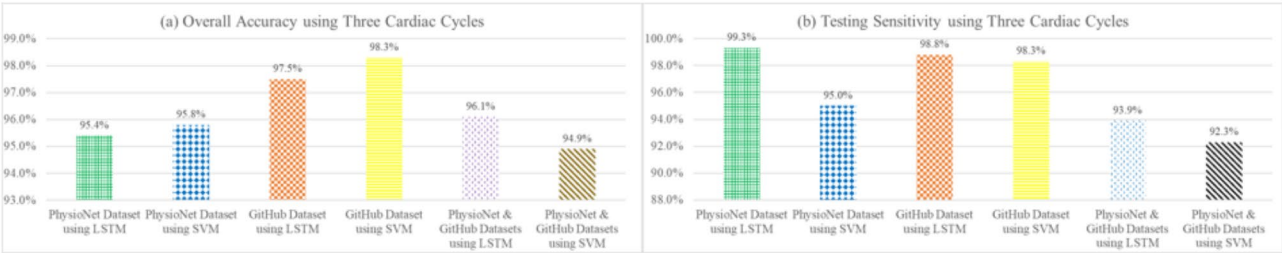

Fig. 12. (**a**) Overall accuracy (**b**) testing sensitivity for normal/abnormal PCG classification based on three cardiac cycles and two feature vectors.

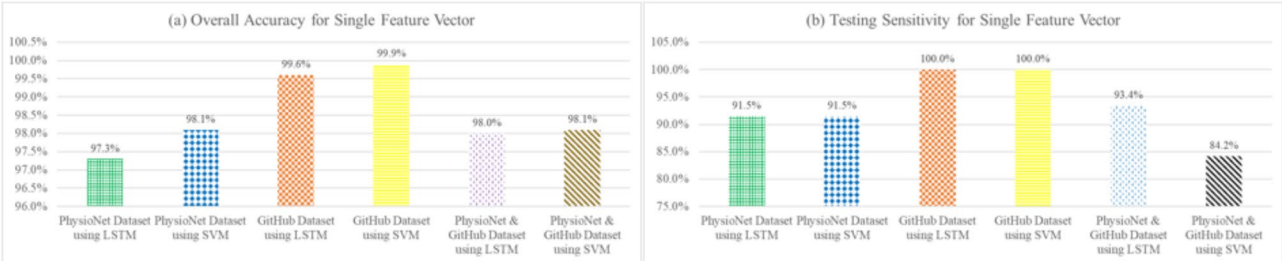

Fig. 13. (**a**) Overall accuracy (b) testing sensitivity for normal/abnormal PCG classification based on a single cardiac cycle and single feature vector.

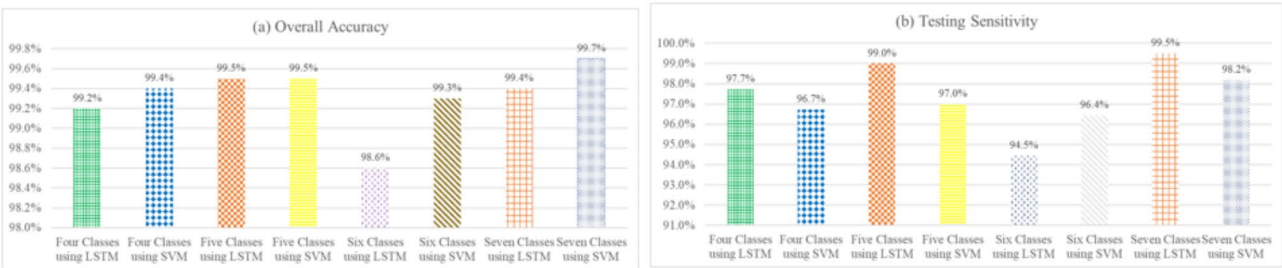

Fig. 14. (**a**) Overall accuracy (**b**) testing sensitivity for abnormal PCG classification based on a single feature vector.

Figure 14 summarizes the overall accuracy and testing sensitivity for Abnormal PCG classification based on a single feature vector.

Tables 6, 7 and 8, and Table 9 demonstrate the testing accuracy for the LSTM and the SVM of by using GitHub and Pascal datasets.

	LSTM (%)	SVM (%)
Testing accuracy	98.4	96.9
Overall accuracy	99.2	99.4
AS Sensitivity	95.2	100.0
MR Sensitivity	100.0	97.7
MS Sensitivity	100.0	93.9
MVP Sensitivity	95.7	95.3

Table 6. Classification results for the four abnormal classes from the GitHub dataset.

	LSTM (%)	SVM (%)
Testing accuracy	99.0	97.5
Overall accuracy	99.5	99.5
AS sensitivity	100.0	100.0
MR sensitivity	100.0	95.7
MS sensitivity	100.0	100.0
MVP sensitivity	96.0	91.7
Overall sensitivity	99.0	97.0
Specificity	100.0	100.0

Table 7. Classification results for the five classes including normal and four abnormal classes from the GitHub dataset.

	LSTM (%)	SVM (%)
Testing accuracy	96.6	96.9
Overall accuracy	98.6	99.3
AS sensitivity	100.0	85.7
MR sensitivity	100.0	100.0
MS sensitivity	100.0	100.0
MVP sensitivity	100.0	100.0
Extrahls sensitivity	66.7	92.9
Extrasystole sensitivity	100.0	100.0

Table 8. Classification results for the six classes from GitHub and PASCAL datasets.

	LSTM (%)	SVM (%)
Testing accuracy	98.5	98.5
Overall accuracy	99.4	99.7
AS sensitivity	100.0	100.0
MR sensitivity	100.0	100.0
MS sensitivity	100.0	100.0
MVP sensitivity	100.0	100.0
Extrahls sensitivity	90.0	87.5
Extrasystole sensitivity	100.0	100.0
Overall sensitivity	99.5	98.2
Specificity	100.0	100.0

Table 9. Classification results for the seven classes including normal and six abnormal classes from the GitHub and PASCAL datasets.

Discussion

Interpretation of key results

The three main issues this research study focuses on are first: segmenting the PCG signal into four regions, S1, systolic, S2, and diastolic segments. Second: recognizing Normal and Abnormal heart sounds in a

timely manner method. Third: classifying the type of detected Abnormality.

For the first research issue, the obtained results for segmenting the PCG signal indicate that the chosen method is timely and efficient by applying simple DSP methods. Also, applying the segmentation method as declared by the obtained results is effective in determining the S1, systolic, S2, and diastolic periods.

By considering the second research issue, our findings suggest that first when obtaining two feature vectors the systolic accuracy is usually higher than the diastolic, regardless of the type of the classifier. Secondly, when applying the same methodology to various combinations of datasets, the GitHub dataset results of taking one cardiac cycle separated into two feature vectors is better than taking one feature vector due to three factors. One factor is the limited number of abnormal classes, the other is the clarity of the input signals, and finally the existence of abnormality in each cardiac cycle. According to the results, using a single classifier provides better efficiency than using two classifiers. For the third research issue, our outcomes from detecting several types of abnormalities reflect the robustness of the applied methodology regardless of the used dataset. Consequently, several experiments showed the ability of the system to identify those abnormalities and even distinguish the Normal classes as shown.

Results comparison

In this section, the results are discussed by comparing them with the existing knowledge on the topic of heart sound classification. In some aspects, it is difficult to conduct an unbiased comparison of our findings and those of other works due in terms of the number of abnormal classes and the utilized datasets. However, for comparison purposes, we applied similar experiments as in the GitHub dataset by adding a Normal class to the detected four abnormal classes. The state-of-art measurements comparison is conducted in terms of overall accuracy, sensitivity, and specificity.

Before continuing, it is important to restate two major contributions of this work. Firstly, to the best of our knowledge, this is the first work that detects six abnormal heart sounds in a timely manner. Also, to the best of our knowledge, the f succeeds to identify S1 and S2 by offering acceptable results with less computational time. The proposed approach shows an enhancement is 0.45 s compared with the mEWT in[16] and with an increase in the total error of 24,084.80 ms for dataset A, and 6,516.2 ms for dataset B.

In a nutshell, we can notice that the proposed approach achieves better accuracy than presented in[17,43,44] approaches using a single SVM. The results in Figs. 15, 16 shows that the proposed approach achieves an accuracy of 96.0% for two SVMs and 98.1% for a single SVM compared to 97.77, 94.07, and 92.23% for[17,43,44] respectively, with a decrease of about 1.77 for two SVMs and an enhancement of about 0.33% for a single SVM.

By applying classical approaches for heart sound recognition, the proposed method can detect six common abnormal sounds, and to the best of our knowledge this is the first work that combines the PASCAL and the GitHub datasets' abnormalities by achieving an overall accuracy of 98.6% using the LSTM classifier, and 99.3% using the SVM classifier.

To compare our results with the state-of-art, an experiment for classifying five classes using the GitHub dataset. The proposed method scored an overall accuracy of 99.5% for the LSTM and the SVM, which is similar to the results found in[15] as shown in (Figs. 15, 16). The results outperform those obtained in[18,45] by 98.9 and 98.5% respectively approaches as shown in (Figs. 15, 16). Taking the MFCC for the DTCWT shows better results compared to other feature extraction methods. Generally, the results show that the LSTM can provide better results in terms of accuracy, sensitivity, and specificity, although it needs more time compared to the SVM. However, we should take into consideration that the proposed system can combine two works, the first one is to detect abnormal PCGs, and the second one is to define the type of detected abnormality. It can detect six, and four abnormal PCGs in a relatively short time, with good accuracy, and sensitivity for each class, also, it can recognize five and seven Normal and abnormal classes. Whereas the just-mentioned works can detect either normal/abnormal PCGs or detect the type of abnormality with a limited number of classes compared to the proposed methodology.

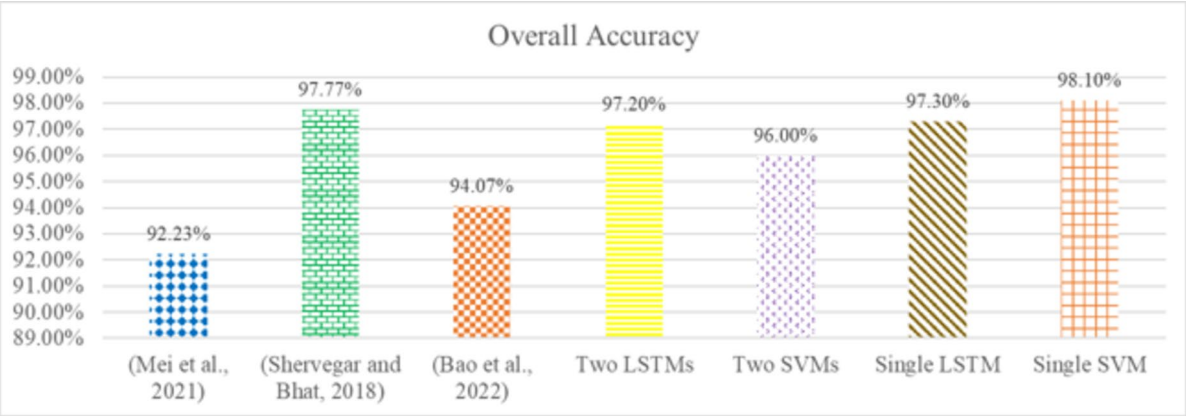

Fig. 15. Comparison between the performance of different approaches for normal/abnormal classification.

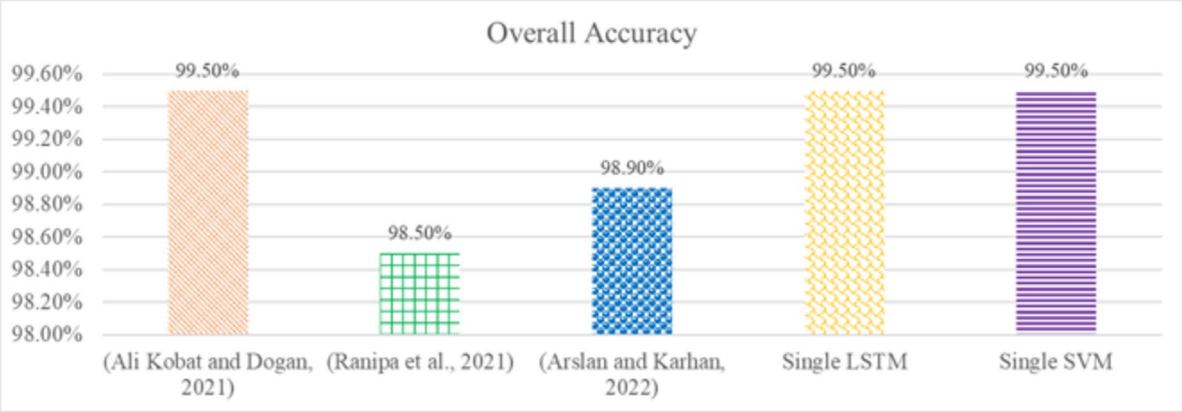

Fig. 16. Comparison between the performance of different approaches for five abnormal classes.

Limitations

The proposed heart sound detection system shows considerable potential, although it possesses inherent limitations that require consideration. A significant difficulty is the inclusion of noise in real-world data, such ambient sounds, motion aberrations, or interference from recording devices, which can distort phonocardiogram (PCG) signals and affect segmentation accuracy. Moreover, inter-patient variability in heart sounds, resulting from variations in age, physiology, and clinical states, presents a challenge to the model's generalizability. We should also mention one limitation of having a relatively low specificity (64.9%) and its implications for false positives, due to the overlap of features and dataset characteristics.

Conclusions

The overall purpose of this current work is to propose a solution for the problem of heart sound recognition in a real-time mode based on examining the heart sound signal segmented features. Based on a quantitative analysis of several performance measures, it can be concluded that the combination of digital signal processing techniques, to segment the input and extract representative features using the Mel-frequency cepstral coefficients for the Dual-tree complex wavelet transform, and machine learning algorithms including the LSTM and the SVM, to recognize the abnormalities, is a valid solution for the research problem. The experimental results indicate that the proposed system can be applied to detect abnormalities and define the type of abnormality effectively in a real-time mode. This research investigates the heart sound segmentation problem and finds a new segmentation method with less computational time. Finally, in the context of other related studies, the obtained results were notable in terms of the targeted measures.

The importance of this work can be seen through its main contributions, which can be summarized as follows:

- A new segmentation method for heart sound signals.
- Original feature extraction in the heart sounds classification method by combining the MFCC with DTCWT.
- Recognizing Normal from Abnormal heart sounds in a real-time mode.
- Classifying six different heart sound abnormalities.
- Presenting a more accurate heart sound abnormality classification system with respect to the present state-of-art work.

Future work

With the increased risk of cardiovascular diseases, the improvements and the growth in the AI field by empowering machines to detect human diseases with good accuracy, and time without the need for experts' presence in their daily life, can enhance community health and reduce mortality. To cover more heart abnormal sounds, find better computational time, and screen more variation among individuals, future studies could address the problem of heart sound classification.

As a part of future work, the intention is to follow up the work in this research with the following directions:

- Collecting datasets including other types of abnormalities and having more individual variations.
- Attempting to explore segmentation approaches for the PCG signal.
- Obtaining less time and more accurate results for detecting abnormal heart sounds opens up several avenues for further research.
- Enhancing the classification model to encompass a wider range of abnormalities, including particular murmurs, arrhythmias, or other uncommon heart diseases, will improve its diagnostic efficacy.

Data availability

The datasets generated and/or analysed during the current study are available in :1-The PhysioNet available at: https://physionet.org/about/database/. 2-PASCAL available at: https://istethoscope.peterjbentley.com/heartchallenge/index.html. 3-Open-Access GitHub datasets are used for Abnormality classification "GitHub - Yaseen21k

han/Classification-of-Heart-Sound-Signal-Using-Multiple-Features-: Data plus Code Fo Classification of Heart Sound Signal Using Multiple Features." available at: https://github.com/yaseen21khan/Classification-of-Heart-Sound-Signal-Using-Multiple-Features-/tree/master.

Received: 2 July 2024; Accepted: 6 February 2025
Published online: 17 March 2025

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Author contributions

Moyawiah AL-Shannaq and Areen Nasrawi conceived the research idea, wrote the paper, and contributed to the analysis and interpretation of the results. Abed Al-Raouf K. Bsoul played a role in reviewing and revising the manuscript. Ahmad Saifan conducted the data analysis and utilized specific tools to analyze the collected data. All authors also contributed to interpreting the results and provided valuable insights throughout the research process.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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